

Least Squares Fitting Functions

Linear Regression

$$y = mx + b \quad m = \frac{y_1 - y_2}{x_1 - x_2}$$

More than 2 pts
no exact sol.

exact sol.

~~| x | y |
|---|---|
| 1 | 1 |
| 2 | 2 |
| 3 | 2 |
| 3 | 3 |~~

Want

$$m(1) + b = 1$$

$$m(2) + b = 2$$

$$m(3) + b = 2$$

$$m(3) + b = 3$$

Not even
a function

$\vec{b}$ unit
length

(Forget interpretation)

$$A = \begin{pmatrix} 1 & 1 \\ 2 & 1 \\ 3 & 1 \\ 3 & 1 \end{pmatrix} \quad x = \begin{pmatrix} m \\ b \end{pmatrix}$$

$$\begin{aligned} m + b &= 1 \\ 2m + b &= 2 \\ 3m + b &= 2 \\ 3m + b &= 3 \end{aligned}$$

$$Ax = b$$

$$b = \begin{pmatrix} 1 \\ 2 \\ 2 \\ 3 \end{pmatrix}$$

Handwritten text in Arabic script, likely a religious or historical document. The text is written in black ink on aged paper. There are several red ink markings, including a large red 'S' or 'Z' shape on the left side and a red signature or stamp on the right side. The text is arranged in several lines, with some words appearing to be in a different script or dialect.

ls + sq (A, B)

Get $X = [0.727, 0.363]$

(Note: In the original image, 'm' is written above the first element and 'b' is written above the second element, with arrows pointing down to them.)

Sol $y = 0.727x + 0.363$

Choose order of parameters
in vector x

Often x in Formula

different from x in $AX=b$



Two years of data

Spring Summer Fall Winter

2012	15	25	16	9
2003	14	22	12	10

Formula: $a \sin t + b \cos t + c$

$\sin t$	0	$\frac{1}{2}$ 0.5	1	-0.5
$\cos t$	1	0.8	0	0.8

Make System = Want

$$a(0) + b(1) + c = 15$$

$$a(0) + b(1) + c = 14$$

$$a(.5) + b(.8) + c = 25$$

$$a(.5) + b(.8) + c = 22$$

$$a(1) + b(0) + c = 16$$

More
eq than var.

No sol.

1. The first part of the paper is devoted to the study of the properties of the function $f(x)$ defined by the equation $f(x) = \int_0^x f(t) dt$. It is shown that $f(x)$ is a constant function.

2. In the second part, we consider the function $g(x)$ defined by the equation $g(x) = \int_0^x g(t) dt$. It is shown that $g(x)$ is a constant function.

3. The third part of the paper is devoted to the study of the properties of the function $h(x)$ defined by the equation $h(x) = \int_0^x h(t) dt$. It is shown that $h(x)$ is a constant function.

4. In the fourth part, we consider the function $k(x)$ defined by the equation $k(x) = \int_0^x k(t) dt$. It is shown that $k(x)$ is a constant function.

5. The fifth part of the paper is devoted to the study of the properties of the function $l(x)$ defined by the equation $l(x) = \int_0^x l(t) dt$. It is shown that $l(x)$ is a constant function.

6. In the sixth part, we consider the function $m(x)$ defined by the equation $m(x) = \int_0^x m(t) dt$. It is shown that $m(x)$ is a constant function.

7. The seventh part of the paper is devoted to the study of the properties of the function $n(x)$ defined by the equation $n(x) = \int_0^x n(t) dt$. It is shown that $n(x)$ is a constant function.

8. In the eighth part, we consider the function $o(x)$ defined by the equation $o(x) = \int_0^x o(t) dt$. It is shown that $o(x)$ is a constant function.

9. The ninth part of the paper is devoted to the study of the properties of the function $p(x)$ defined by the equation $p(x) = \int_0^x p(t) dt$. It is shown that $p(x)$ is a constant function.

10. In the tenth part, we consider the function $q(x)$ defined by the equation $q(x) = \int_0^x q(t) dt$. It is shown that $q(x)$ is a constant function.

11. The eleventh part of the paper is devoted to the study of the properties of the function $r(x)$ defined by the equation $r(x) = \int_0^x r(t) dt$. It is shown that $r(x)$ is a constant function.

12. In the twelfth part, we consider the function $s(x)$ defined by the equation $s(x) = \int_0^x s(t) dt$. It is shown that $s(x)$ is a constant function.

13. The thirteenth part of the paper is devoted to the study of the properties of the function $t(x)$ defined by the equation $t(x) = \int_0^x t(t) dt$. It is shown that $t(x)$ is a constant function.

14. In the fourteenth part, we consider the function $u(x)$ defined by the equation $u(x) = \int_0^x u(t) dt$. It is shown that $u(x)$ is a constant function.

15. The fifteenth part of the paper is devoted to the study of the properties of the function $v(x)$ defined by the equation $v(x) = \int_0^x v(t) dt$. It is shown that $v(x)$ is a constant function.

16. The sixteenth part of the paper is devoted to the study of the properties of the function $w(x)$ defined by the equation $w(x) = \int_0^x w(t) dt$. It is shown that $w(x)$ is a constant function.

17. In the seventeenth part, we consider the function $x(x)$ defined by the equation $x(x) = \int_0^x x(t) dt$. It is shown that $x(x)$ is a constant function.

18. The eighteenth part of the paper is devoted to the study of the properties of the function $y(x)$ defined by the equation $y(x) = \int_0^x y(t) dt$. It is shown that $y(x)$ is a constant function.

19. In the nineteenth part, we consider the function $z(x)$ defined by the equation $z(x) = \int_0^x z(t) dt$. It is shown that $z(x)$ is a constant function.

20. The twentieth part of the paper is devoted to the study of the properties of the function $a(x)$ defined by the equation $a(x) = \int_0^x a(t) dt$. It is shown that $a(x)$ is a constant function.

21. The twenty-first part of the paper is devoted to the study of the properties of the function $b(x)$ defined by the equation $b(x) = \int_0^x b(t) dt$. It is shown that $b(x)$ is a constant function.

22. In the twenty-second part, we consider the function $c(x)$ defined by the equation $c(x) = \int_0^x c(t) dt$. It is shown that $c(x)$ is a constant function.

23. The twenty-third part of the paper is devoted to the study of the properties of the function $d(x)$ defined by the equation $d(x) = \int_0^x d(t) dt$. It is shown that $d(x)$ is a constant function.

24. In the twenty-fourth part, we consider the function $e(x)$ defined by the equation $e(x) = \int_0^x e(t) dt$. It is shown that $e(x)$ is a constant function.

25. The twenty-fifth part of the paper is devoted to the study of the properties of the function $f(x)$ defined by the equation $f(x) = \int_0^x f(t) dt$. It is shown that $f(x)$ is a constant function.

26. The twenty-sixth part of the paper is devoted to the study of the properties of the function $g(x)$ defined by the equation $g(x) = \int_0^x g(t) dt$. It is shown that $g(x)$ is a constant function.

27. In the twenty-seventh part, we consider the function $h(x)$ defined by the equation $h(x) = \int_0^x h(t) dt$. It is shown that $h(x)$ is a constant function.

28. The twenty-eighth part of the paper is devoted to the study of the properties of the function $i(x)$ defined by the equation $i(x) = \int_0^x i(t) dt$. It is shown that $i(x)$ is a constant function.

29. In the twenty-ninth part, we consider the function $j(x)$ defined by the equation $j(x) = \int_0^x j(t) dt$. It is shown that $j(x)$ is a constant function.

30. The thirtieth part of the paper is devoted to the study of the properties of the function $k(x)$ defined by the equation $k(x) = \int_0^x k(t) dt$. It is shown that $k(x)$ is a constant function.

$$A = \begin{pmatrix} 0 & 1 & 1 \\ 0 & 1 & 1 \\ .5 & .8 & 1 \\ .5 & .8 & 1 \\ 1 & 0 & 1 \\ \vdots & \vdots & \vdots \end{pmatrix} \quad b = \begin{pmatrix} 15 \\ 14 \\ 25 \\ 22 \\ 16 \\ \vdots \\ \vdots \\ \vdots \end{pmatrix}$$

$$X = \begin{pmatrix} a \\ b \\ c \end{pmatrix}$$

Make a function for rows of A ... b

$$\text{Row of } A = (\sin t_i, \cos t_i, 1)$$

Function of t

Function of t

Row of $b = y_i$

t_i, y_i are in data

